Paleomagnetic Study and Chemistry of Newer Dolerites from Singhbhum, Bihar, India

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Results of paleomagnetic studies on seventeen dikes sampled from the Newer Dolerites of Singhbhum, between latitude 22°0′N to 22°45′N and longitude 85°48′E to 86°25′E are reported. The dikes are of Precambrian age, probably older than 1000 my. Stable directions of magnetization were obtained for eleven sites by thermal demagnetization in a field free space in temperatures ranging from 250° to 400°C. These studies showed that the dikes had been reheated by igneous activity of neighboring dikes of a younger age or by regional metamorphism of a younger period. Polished section studies carried out on these dikes showed that the magnetic mineral had been significantly altered as a result of these processes. Reliable virtual geomagnetic poles were obtained for ten different sites. These pole positions can be collected into three different groups which may represent three episodes of igneous activity of the Newer Dolerites. A very tentative polar wandering curve for the Precambrian in India is given.

Nous rapportons ici les résultats d'études paléomagnétiques faites sur dix-sept dykes échantillonnés dans les Nouvelles Dolérites de Singhbhum, entre les latitudes 22°0′N et 22°45′N et les longitudes 85°48′E et 86°25′E. Ces dykes, d'âge précambrien, ont probablement plus de 1000 millions d'années. Des directions stables de magnétisation furent obtenues sur onze sites par demagnétisation thermale dans un espace libre de champ à des températures variant entre 250° et 400°C. Ces études montrent que les dykes ont été réchauffés par l'activité ignée des dykes voisins plus jeunes ou par un métamorphisme régional plus récent. Comme aboutissement de ces processus, les minéraux magnétiques ont été altérés de façon très significative, c'est ce qu'a fait ressortir l'étude de sections polies. Nous avons obtenus des pôles géomagnétiques virtuels reliables en dix sites différents. Ces positions de pôles peuvent être distribuées en trois groupes qui peuvent représenter trois épisodes d'activité ignée des Nouvelles Dolérites. Nous avons essayé d'établir une courbe du déplacement polaire pour le Précambrien en Inde.
[Traduit par le journal]

Introduction

Paleomagnetism is an important tool for geological correlation of igneous as well as sedimentary rocks. In India such geological correlation has been attempted for dolerite dikes from Gondwana basins of Damodar Valley (Athavale and Verma 1970). Husnain and Qureshy (1971) have studied several dikes from the southern peninsular shield of India. On the basis of paleomagnetic studies, some of the dikes have been found to be correlatable with the Deccan traps. From other countries there are examples of classification of dikes on the basis of paleomagnetic studies, such as Evans (1968) in Australia, Strangway (1965, 1966) and Larson and Strangway (1969) in Canada and the U.S.A. In this paper we report results of paleomagnetic studies of the Newer Dolerites from Singhbhum, Bihar, India.

During the Precambrian there appear to have been three main episodes of dike intrusion throughout peninsular India (Pascoe 1950, p. 469). The earliest corresponded to the contemporaneous traps of the Dharwar system. This was uniformly felt over the whole of peninsular India. The second episode is represented by dikes and sills of the Cuddapah system. This was recognized mainly in many parts of the southern peninsula but more restrictedly in the northern half. The third one coincided with the outbursts of Malani rhyolites. This is more evident in the northern half of peninsula.

The volcanic types of the Dharwar and those of Cuddapah are in most cases sharply contrasted by the degree of alteration they exhibit.

TABLE 1. Stratigraphic sequence in the Singhbhum area

According to Sarkar *et al.* (1969)	According to Dunn and Dey (1942)
(End of Singhbhum cycles 850 my)	Newer Dolerites
Biotite granite gneiss	Granite
Newer Granophyres	Ultrabasic rocks
Dolerites Gabbro-anorthosite	Basic lavas
Ultramafic intrusions	Upper Shales
Kolhan group	Banded Hematite
Jagannathpur lavas	Quartzite
Unconformity	Lower Shales
End of iron ore cycle, 2700 my	Purple Limestone
Singhbhum granite	Basal Conglomerate
Epidiorites	Older Metamorphics
Iron ore group	
Unconformity	
End of older metamorphic cycle 3200 my	
Older metamorphic group	

The former are mostly hornblende plagioclase rocks, showing marked changes acquired during metamorphism of normal dolerites into epidiorites and hornblende schists. The Cuddapah lavas are on the other hand more free from signs of regional metamorphism. The dikes belonging to the second category are believed to be fairly wide spread in shield areas of Andhra Pradesh, Mysore, Madras, Madhya Pradesh, and Bihar. The exact ages of eruption of these dikes being unknown, paleomagnetic methods can be used for the correlation of igneous activity from these areas although these are fairly wide spread in distance from each other.

The Newer Dolerites in the Singhbhum district of Bihar, India form a very impressive volcanic formation. These dolerites are mostly in the form of dikes spread all through the Singhbhum granite and the adjoining areas. Paleomagnetic studies on several dikes from the northern part of Singhbhum have been carried out with a view to study, (a) the history of igneous activity in the area, (b) to attempt classification of the dikes on the basis of their mean directions, pole positions, or magnetic properties, and (c) to obtain pole positions for the period corresponding to the age of the Newer Dolerites and to provide a basis for correlation of igneous activity of Singhbhum with that of other parts of India such as the Cuddapah basin.

Geology of the Area

Geologically Singhbhum is an area of very complex geology in Bihar State of India. Here sedimentary, volcanic, and metamorphic formations are found ranging in age from about 3500 my to about 850 my (Sarkar 1968). Predominent formations in the area are metamorphics of various grades, iron ore series, Dalma and Dhanjori volcanics, Singhbhum granite, Newer Dolerites, and Soda granite. According to Sarkar *et al.* (1969), Singhbhum has undergone three distinct orogenic cycles accompanied by granitic activity and closing at 3200, 2700, and 850 my. Adjoining areas of Singhbhum have also suffered regional metamorphism of 890–970 my (Sarkar 1968).

The geology of Singhbhum and adjacent areas has been studied in detail by Jones (1934), Dunn (1937), Dunn and Dey (1942), and Sarkar *et al.* (1969). Dunn and Dey (1942), in their classification, assigned the youngest age to the Newer Dolerites. Recently however, Sarkar *et al.* (1969) have revised the stratigraphy of Singhbhum and attributed a wide span of age to this volcanic activity. Sarkar (1968) points out that the basic intrusions of the Newer Dolerite suite continued intermittently, the youngest being probably around 1000 my old. The stratigraphic sequence in this area is as shown in Table 1. Sarkar *et al.* (1969) have obtained K–Ar radiometric ages of two dikes from the southern part of Singhbhum at 1540 and 1690 my. This age of igneous activity partly coincides with that of Cuddapah basin in the south. Dunn (1937) states that "the last great series of igneous intrusions, the Newer Dolerites, probably of Cuddapah age but more probably Archean,

gave rise to enormous dikes which cut the granite in the form of a very regular network with two principle strike directions almost at right angles to each other". The two principal strike directions are NNW–SSE and NNE–SSW. The dikes are often several kilometers in length and are seen to cut across the Singhbhum granite, adjoining the iron ore series and older metamorphics. The thickness varies from a few meters to 700 meters. The dolerites are variable in texture; some are fine grained, the majority are medium grained, and few are coarse grained. The principal minerals are augite and labradorite. Some of the augite is titaniferous. The dikes are slightly metamorphosed. A geological map of part of the area where samples were collected, is shown in Fig. 1.

Sampling

The Newer Dolerites traversing the Singhbhum granite between latitude 22°0′N to 22°45′N and longitude 85°48′E to 86°25′E were sampled at seventeen different locations. A minimum of 5–8 hand oriented blocks were taken from each site. The samples were collected from road cuttings, railway cuttings, river exposures, and quarries. Locations of some of the dikes sampled are shown in Fig. 1.

Laboratory Studies

NRM Susceptibility, and Q_N Ratio

Out of each sample, 3–4 specimens of 1 in. diameter were measured using an astatic magnetometer of sensitivity 3.6×10^{-6} Oe/cm deflection, at the laboratory of the Indian School of Mines, Dhanbad, India. The magnetic susceptibilities of the dikes were determined at the National Geophysical Research Institute, Hyderabad, using an apparatus similar to that of Likhite and Radhakrishnamurty (1965), in a field of 1.0 Oe. The Q_N ratio was determined for each specimen using a 0.5 Oe value for the intensity of the earth's magnetic field. The variations in NRM intensity, susceptibility, and Q_N ratio for all the dikes are given in Table 2. It is noted that the NRM intensities are mostly in the range 0.2×10^{-4} to 25×10^{-4} Gauss, susceptibility in the range 0.4×10^{-4}–1.0×10^{-4} Gauss/Oe and Q_N ratios in the range 0–10. These susceptibilities as well as NRM intensities are much lower than those of fresh dolerites, such as the ones encountered in the Gondwana basins of Damodar Valley in Bihar (Athavale and Verma 1970). The NRM directions from some of the sites are shown in Figs. 2a and 2b. These show a considerable scatter indicating that the rocks have been much affected by secondary components of magnetization.

Stability Tests

Stability tests using alternating field demagnetization techniques were applied using apparatus similar to that of Creer (1959). This technique was successful only in removing unstable components from A and Q sites. Figures 3a and 3b show changes in the direction of magnetization on progressive demagnetization in alternating fields up to 300 Oe for site A and 200 Oe for site Q. AF demagnetization curves of normalized intensity for the specimen are also shown in these figures. For some sites af demagnetization partly removed unstable components, while for others scatter in direction increased considerably on using this technique.

As af demagnetization was not successful in magnetic cleaning, thermal demagnetization was tried for all dikes in progressively increasing temperatures of 50, 100, 200, 300, and 400 °C above room temperature. The thermal demagnetization apparatus constructed at N.G.R.I. (Athvale 1969) was used for this purpose. Typical examples of change in magnetic vector on thermal cleaning are shown in Figs. 4a and 4b for sites C and I. From other sites the stable directions were obtained after thermal demagnetization in temperatures ranging from 250 to 400 °C as shown in Fig. 4c. Mean direction of magnetization for ten different dikes after af and thermal demagnetization are given in Table 3, along with other paleomagnetic parameters including pole positions. Rocks from remaining dikes could not be cleaned satisfactorily using mentioned techniques.

Magnetic Properties

In order to study the nature of the magnetic minerals present, other magnetic properties were investigated apart from susceptibility as mentioned earlier. These included: (1) saturating field, (2) remanent coercivity, and (3) Curie temperature.

The first two were determined by plotting

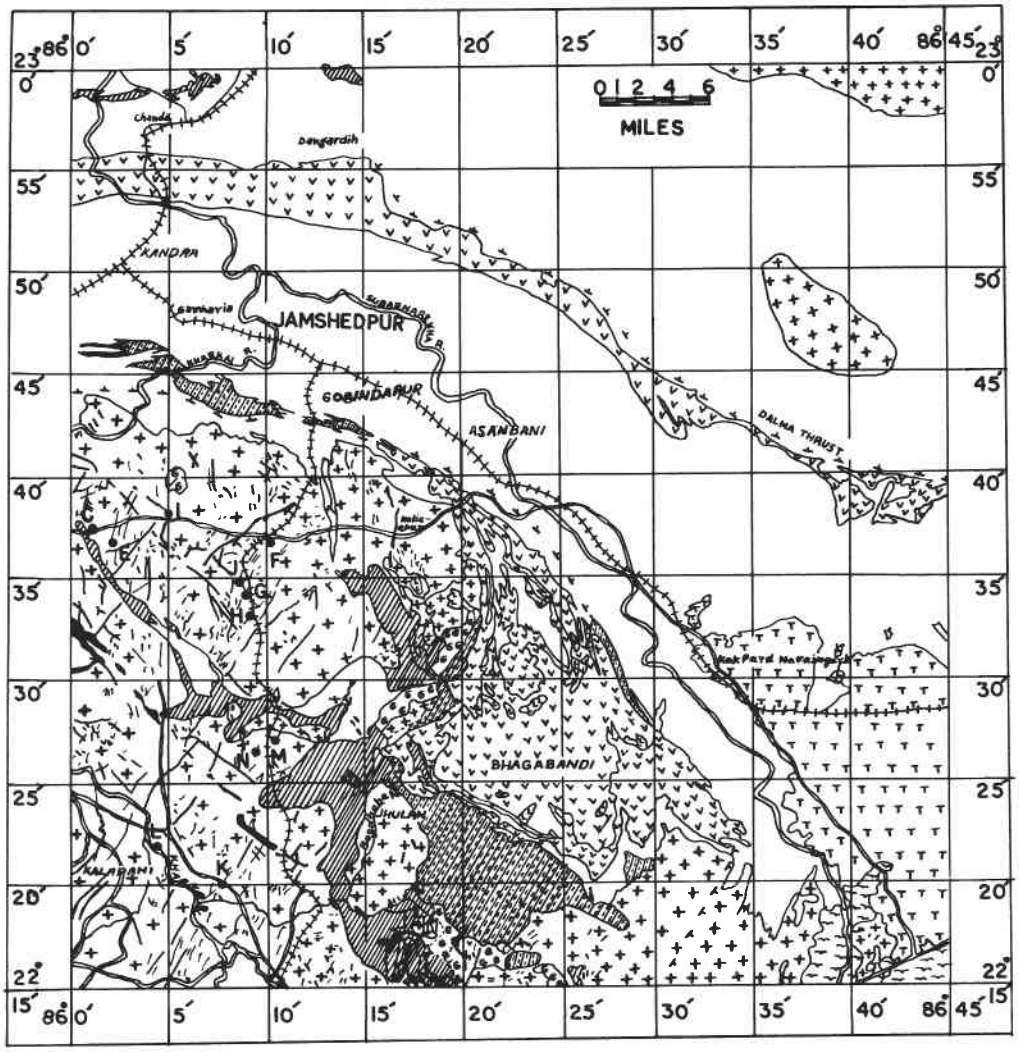


FIG. 1. Geological map of part of eastern Singhbhum showing the location of some of the dikes sampled (prepared by J. A. Dunn, A. K. Dey, and A. B. Dutt).

TABLE 2. Magnetic properties of various dikes from Singhbhum

Site	Susceptibility (Gauss/Oe) Range ($\times 10^{-6}$)	Susceptibility (Gauss/Oe) Mean ($\times 10^{-6}$)	Q_N ratio Range	Q_N ratio Mean	Saturation intensity (10^{-3} emu/cc)	Saturation intensity (emu/gm)	Remanent coercivity (H_c) (Oe)	Saturating field (Oe)	Curie temp (°C)
A	36–89	69.44	0.5–13.4	4.5	2.02	21.6	300	1500	582
C	74–2560	432.18	19.08–119.08	53.06	45.3	52.9	300	2500	578
F	102–114	107.72	4.48–48.71	18.90	—	135.5	—	—	588
H	48.64–70.53	57.62	6.77–114.82	28.00	2.93	96	300	2500	588
I	59.4–167.8	113.2	12.6–84.16	50.79	6.06	102.8	150	1500	590
K	87.16–94.57	90.7	14.58–48.08	34.3	23.5	46	330	2500	591
L	91.25–106.71	95.15	1.62–12.10	5.70	17.5	134.0	575	2500	589
N	89.93–104.24	97.2	0.24–1.30	0.50	5.56	145.0	375	2500	582
O	53.92–94.55	66.86	0.07–2.78	1.52	1.944	34.9	325	2000	601
Q	75.48–107.93	92.4	0.02–29	5.8	1.04	45.5	300	1500	568
B	65.58–88.48	75.00	2.7–41.2	16.7	16.85	82.7	250	2000	576
D	73.85–95.4	81.6	8.0–12	10.00	0.98	142	300	1500	588
E	95.11–186.48	124.75	7.6–8.0	42.33	3.77	112.5	500	1500	599
G	51.5–96.7	81.4	5.9–36.9	25.20	31.7	172	400	2000	601
M	86.95–156.00	99.50	30.4–12.36	7.00	3.81	131.00	1000	3500	593
P	81.33–1447.00	275.6	0.4–66.5	33.00	459.4	29.2	575	2000	356 601

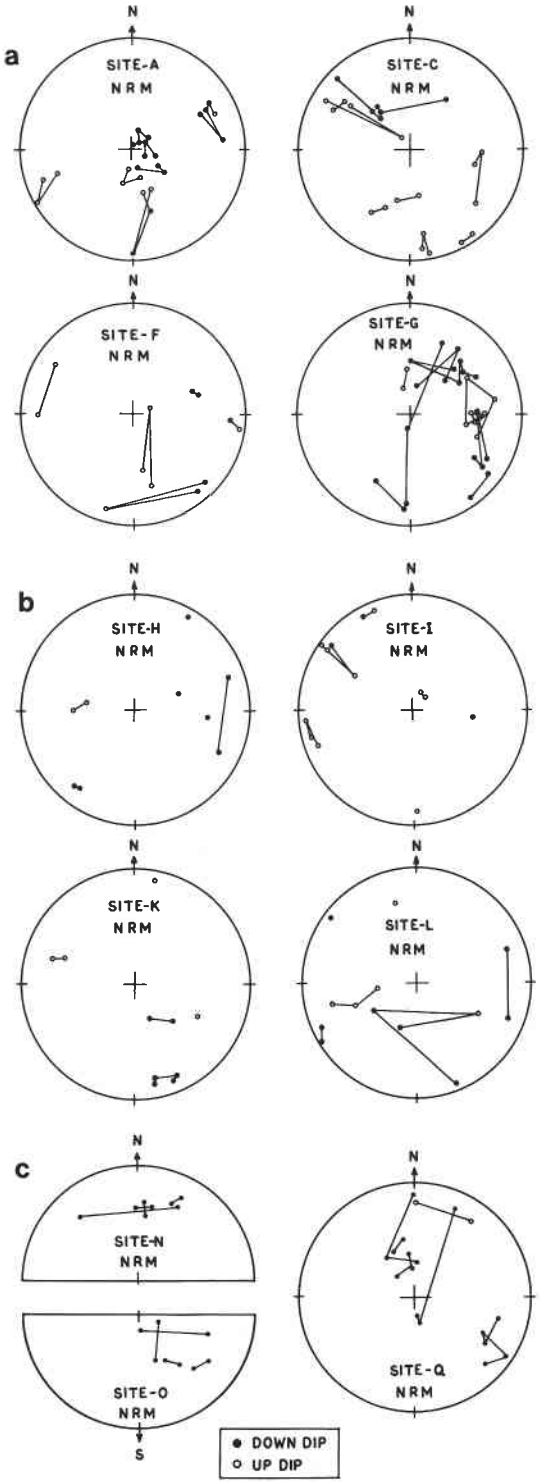


FIG. 2 a, b, and c. NRM directions of some of the sites showing considerable scatter in their directions.

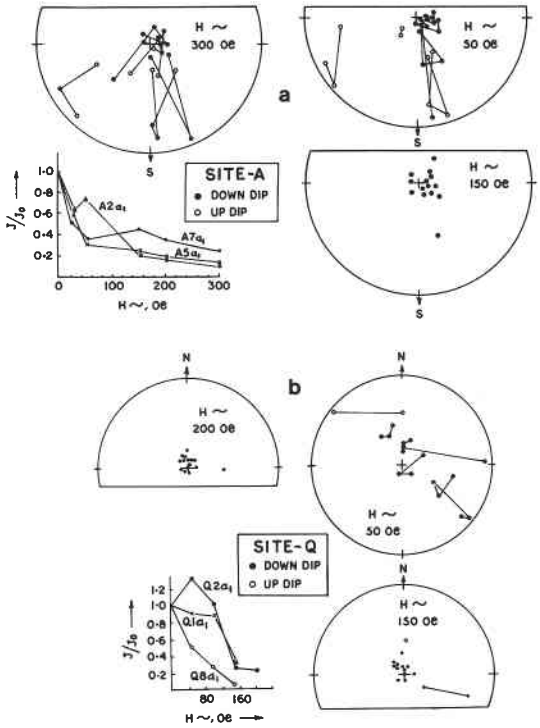
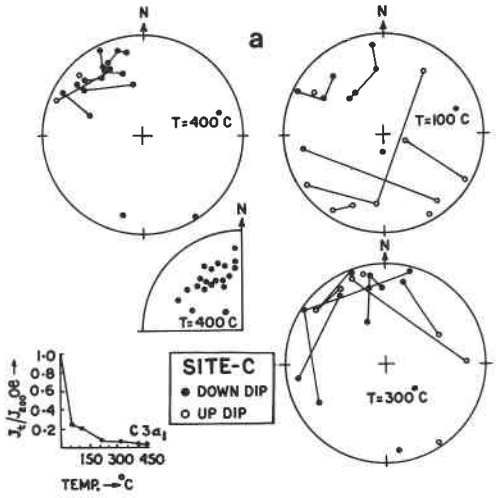


FIG. 3 a and b. Changes in the direction of remanent vector on alternating field demagnetization with progressive field strength for site A and site Q respectively.

intensity vs. field strength in a manner similar to that of Carmichael (1961). The magnetic properties for all the dikes are summarized in Table 2. The saturation intensity varied considerably, from 29.2 emu/gm to 145.0 emu/gm. The remanent coercivity varied from 150 Oe to nearly 1000 Oe and Curie temperatures for all sites were close to that of magnetite, except for site P, which showed two Curie temperatures. The examples of J vs. H curves for determining saturating field, saturation intensity, and coercivity are shown in Fig. 5. A few typical examples of J_s vs. T curves are shown in Fig. 6. On an average, the samples were saturated in fields varying from 1250 to 3500 Oe. The coercivity varied in general from 300 to 500 Oe. For one site (M) it was as much as 1000 Oe. This could be either due to the presence of hematite, as suggested by studies of polished sections, or to the very small size of magnetite grains. Fine grained magnetite particles are known to have a coercivity as high as 3000 Oe.



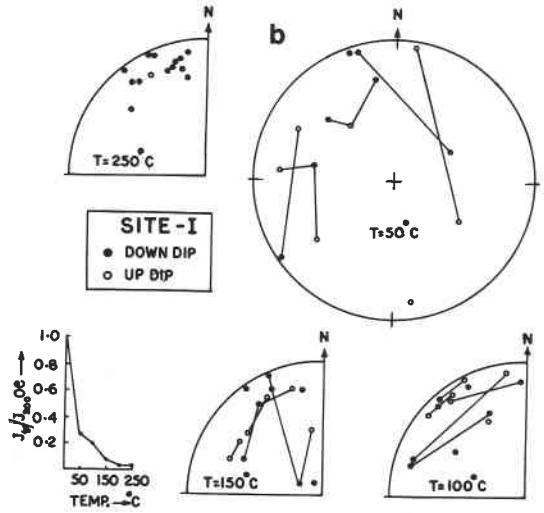


FIG. 4 a and b. Changes in magnetic vector on thermal cleaning at progressively higher temperatures for sites C and I respectively.

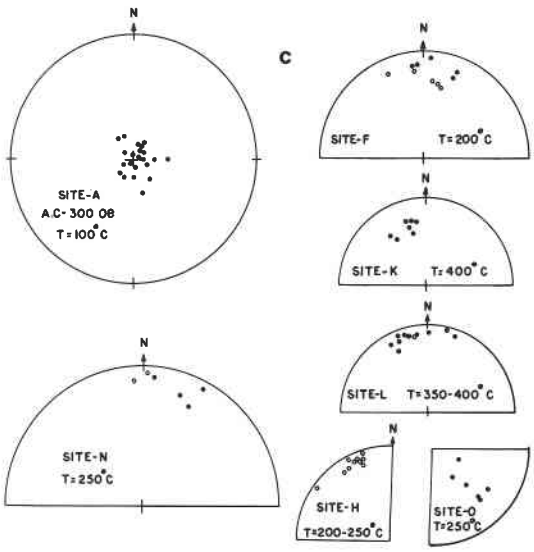


FIG. 4 c. Stable directions of magnetization for sites A, F, H, K, L, N, and O after thermal cleaning in temperatures ranging from 250° to 400 °C.

Petrographic Studies

Petrographic studies were carried out on one sample from each dike. The studies included examination of thin sections, as well as polished sections. The thin section studies showed that the different sites had undergone slight metamorphism. Most of the magnetic mineral had been altered to leucoxene. Only a small part remained unaltered. This explains the low intensity of NRM of most of the dikes. The polished sections studies showed that magnetic grains had been altered and martitization had taken place to different degrees in many of the dikes. Results of alteration of magnetic minerals for two typical dikes are shown in Figs. 7a and 7b.

Chemical analysis of major elements for one sample from each dike are given in Table 4.

An alkali–silica ratio diagram for various dikes is shown in Fig. 8. This shows that the overall composition of the dikes was intermediate between that of tholeitic and alkali dolerites.

Discussion of Results

Results of stability tests using thermal demagnetization on various dikes revealed that the magnetic minerals in the dikes could have been affected by heating in the range of 250° to 400 °C subsequent to the formation of the dikes. The igneous activity represented by the dikes could have lasted for a fairly long period of time. As the dikes are fairly close to each other, the older ones could have been affected by heating from the younger, adjacent dikes. Another possibility is that the dikes could have been affected by a low grade regional metamorphism of a younger age. Alternatively, viscous PTRM could have been important in changing the magnetic history of rocks. This could explain the oxidation of the magnetic minerals, scatter in the NRM directions, and the difficulties encountered in using the alter-

TABLE 3. Paleomagnetic data for all dikes of Singhbhum showing stable directions of remanent magnetization

Site	Treatment	No. of specimens (n)	No. of samples (N)	R	D°	I°	α_{95}°	k	Pole position λ_p	Pole position L_p
C	450°C	19	9	8.6799	326	+23	10	25.0	56°N	16°W
F	200°C	10	5	4.6737	1	−5	21	14.4	65°N	94°W
H	200°C	10	8	7.8032	330	−9	7	66	56°N	52°47′W
I	250°C	14	7	6.8418	335	+13	10	38	61°8′N	34°12′W
K	400°C	7	4	3.9437	334	+30	12	60	65°N	9°W
L	350°C	11	5	4.8761	348	+10	13	32.4	69°N	58°42′W
N	250°C	6	4	3.9025	8	+2	14	44	67°8′N	107°W
O	200°C	6	4	3.8042	142	+46	13	15.32	28°N	56°W
Q	150, 200 Oe	15	6	5.8843	23	+82	10	43.2	36°46′N	94°E
A	300 Oe and 100°C	25	7	6.9289	92	+86	6	85	22°N	94°E

NOTES: R—Resultant of N unit vectors.
α_{95}—Radius of circle of confidence at 95% probability.
k—Precision parameter after Fisher (1953)
λ_p, L_p—Latitude and longitude of virtual geomagnetic pole.
d_p, d_m—Radii of ovals of confidence of pole position.
D—Mean declination of site.
I—Mean inclination of site.

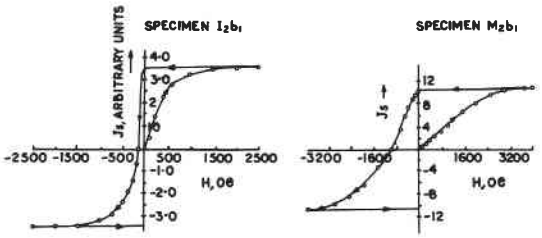


FIG. 5. Two examples of J vs. H curves showing large variation in remanent coercivity and saturating field for different sites.

nating field demagnetization technique for cleaning. Once the effects of subsequent heating (or PTRM) were removed by thermal demagnetization in a field free space, the original directions of magnetization were revealed.

The virtual geomagnetic poles corresponding to the stable directions from various dikes are listed in Table 3.

Pole positions for sites C, F, H, I, K, L, and N are quite close to each other and form one group. The mean pole position (north pole) for this group is 67°N, 46°W (Pole N.D. 1). Poles for sites A and Q possibly belong to another group. The mean pole for these two sides is 29.5°N, 94°E (Pole N.D. 2). The pole position for site O (Pole N.D. 3) is quite different from the other two groups. It therefore appears that there have been at least three different episodes of igneous activity represented by the Newer Dolerites in Singhbhum. The pole positions for three different groups (Poles N.D. 1, N.D. 2, and N.D. 3) are plotted in Fig. 9.

For the Precambrian period in India very few reliable pole positions have been obtained so far. These are listed in Table 5 and plotted in Fig. 9.

On the basis of paleomagnetic data, a limited stratigraphic correlation may be attempted. The pole position N.D. 1 falls between pole 4 for the Kaimur quartzites (910–940 my), and pole 5 and 6 corresponding to the Cuddapah formations (1100–1600 my). The igneous activity represented by this group (sites C, H, F, I, K, L, and N) could therefore be intermediate in age between these two formations.

Large differences in pole positions between three different groups could mean that the igneous activity represented by the Newer Dolerites might have lasted a fairly long period of time and therefore support the stratigraphic sequence given by Sarkar *et al.* (1969) for the Singhbhum region.

It is hoped that these results will be useful in building a Precambrian paleomagnetic stratigraphy for the Indian subcontinent.

ATHAVALE, R. N. 1969. An apparatus for thermal demagnetization of rocks. Bull. N.G.R.I., 2–3, pp. 87–98.

ATHAVALE, R. N. and VERMA, R. K. 1970. Paleomagnetic results from Gondwana dykes from Damodar Valley coal fields and their bearing on the sequence of Mesozoic igneous activity in India. Geophys. J. R. astr. Soc., **20**, pp. 303–316.

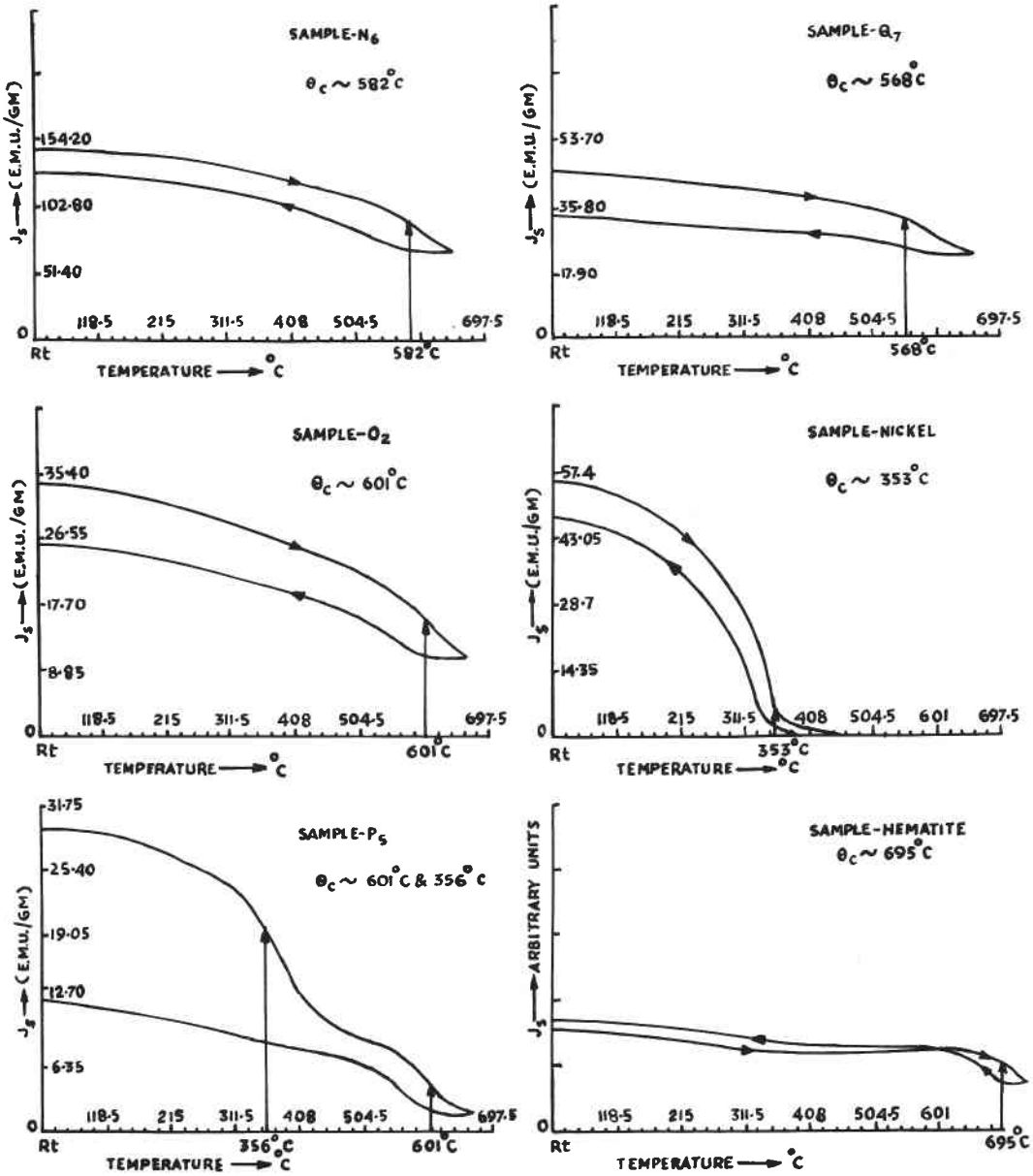


FIG. 6. Typical examples of J_s vs. T curves from different sites. J_s–T curves for standard nickel specimen and hematite ore are also shown.

ATHAVALE, R. N., RADHAKRISHNAMURTY, C., and SAHSRABUDHE, P. W. 1963. Paleomagnetism of some Indian rocks. Geophys. J. R. astr. Soc., 7, pp. 304–313.

ATHAVALE, R. N., ASHA HANSRAJ and VERMA, R. K. 1972. Paleomagnetism and age of Bhander and Rewa sandstones from India. Geophys. J. R. astr. Soc., **98**, pp. 499–509.

ATHAVALE, R. N., VERMA, R. K., BHALLA, M. S., and PULLAIAH, G., 1970. Drift of the Indian sub-continent since Precambrian times. Paleogeophysics, Academic Press, pp. 281–305.

CARMICHAEL, C. M. 1961. The magnetic properties of ilmenite–hematite crystals. Proc. R. Soc. (London), **163**, 508 p.

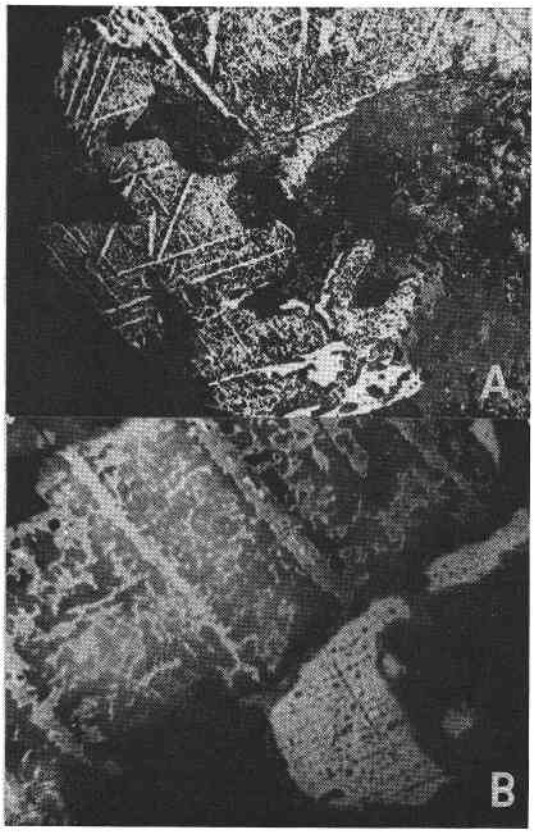


FIG. 7 a and b. Examples of alteration of magnetic minerals as seen in polished sections. (a)—site M, 12.5 × 8 mag. under oil, (b)—site G, 25 × 8 mag. under oil.

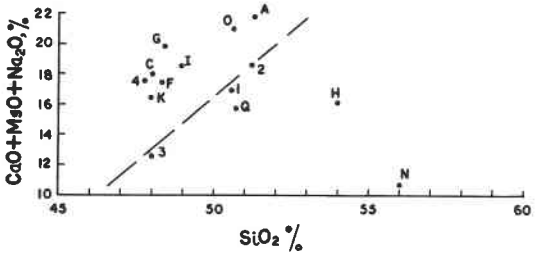


FIG. 8. Alkali–silica ratio diagram for one sample from each dike, showing that the overall compositions of the dikes were intermediate between that of tholeitic and alkali dolerites.

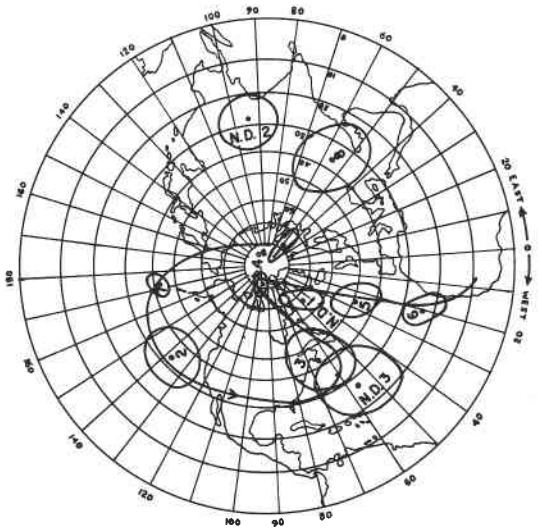


FIG. 9. Pole positions for ten sites of Singhbhum dikes as given in Table 3 and eight Precambrian formations listed in Table 5.

CRAWFORD, A. R. 1969. India, Cylone and Pakistan, new age data and comparisons with India. Nature, **223**, pp. 380–381.

CRAWFORD, A. R. and COMPSTON, W. 1970. The age of Vindhyan System of Peninsular India, Quart. J. Geol. Soc. Lond., **125**, pp. 351–372.

CREER, K. M. 1959. A.C. demagnetization of unstable Triassic Keuper Marls from Southwest England. Geophys. J. astr. Soc., **2**, 261 p.

DUNN, J. A. 1937. Mineral deposits of Eastern Singhbhum and surrounding areas. Mem. Geol. Surv. India, 69(1), pp. 1–280.

DUNN, J. A. and DEY, A. K. 1942. Geology and petrology of Eastern Singhbhum and surrounding areas. Mem. Geol. Surv. India, 69(2), pp. 281–456.

EVANS, M. E. 1968. Magnetization of dikes—A study of the paleomagnetism of Widgiemorltha dike suite, western Australia. J. Geophys. Res., **73**, pp. 3261–3270.

HASNAIN, I. and QURESHY, M. N. 1971. Paleomagnetism and geochemistry of some dykes, Mysore state, India. J. Geophy. Res., **76** (20), pp. 4786–4795.

JONES, H. C. 1934. The iron ores of Bihar and Orissa. Mem. Geol. Surv. India. 63(2), pp. 167–302.

LARSON, E. E. and STRANGWAY, D. W. 1969. Magnetization of Spanish peak dike swarms, Colorado and Shiprock dike, New Mexico. J. Geophys. Res., **74**, pp. 1505–1514.

LIKHITE, S. D. and RADHAKRISHNAMURTY, C. 1965. Apparatus for the determination of susceptibility of rocks in low fields at different frequencies. Bull. N.G.R.I., 3, 1, pp. 1–8.

MANSON, V. 1967. Basalts, The Poldervaart treatise on rocks of basaltic composition. H. H. Hess. and A. Poldervaart, (*Eds.*), Interscience.

MISHRA, D. C. 1965. Unpubl. Ph.D. thesis, Banaras Hindu University, Banaras, India.

PASCOE, EDWIN. 1950. A manual of the Geology of India and Burma, Gov. India Press, Calcutta, **1**.

PRASAD, C. V. R. K. 1966. Unpubl. Ph.D. thesis, Sri Venkateshwara University, Tirupati, India.

TABLE 4. Chemical composition of the Newer Dolerites and others

Component (percentage)	Newer Dolerite samples													
	A	C	F	G	H	I	K	L	N	O	Q	1	2	3
SiO_2	51.34	48.01	48.28	48.42	54.12	48.82	47.98	47.78	56.26	50.64	50.68	50.61	51.20	48.20
Al_2O_3	14.90	14.00	13.94	15.94	14.41	14.04	15.35	14.35	10.90	11.64	14.26	13.58	15.10	16.00
TiO_2	0.54	1.54	1.82	1.18	1.00	1.25	2.00	2.23	1.60	0.91	2.08	1.91	1.50	1.80
Fe_2O_3	2.58	3.50	3.84	2.57	3.04	2.90	3.53	3.47	4.61	1.94	3.75	3.91	2.3	2.8
FeO	5.52	11.40	10.47	9.63	6.94	10.20	10.47	11.04	11.19	8.35	9.06	9.92	8.9	8.3
MnO	0.07	0.12	0.05	0.06	0.03	0.08	0.05	0.07	0.14	0.05	0.10	0.16	0.19	0.17
CaO	11.17	10.35	9.63	11.33	8.58	10.84	9.46	10.05	6.66	7.59	8.61	9.45	10.10	2.8
MgO	8.46	5.77	5.55	6.88	5.18	6.06	4.71	5.30	1.62	11.18	4.21	5.46	6.40	6.50
Na_2O	2.14	1.82	2.14	1.72	2.50	1.75	2.35	2.23	2.52	2.23	2.98	2.60	2.20	3.30
K_2O	0.75	0.64	1.11	0.69	1.61	0.97	1.28	0.78	1.66	1.58	0.92	0.72	0.90	1.30

1. The average of Deccan Trap, after Washington (1922).
2. Averages of all tholeitic dolerites, after Manson (1967).
3. Averages of all alkalic dolerites, after Manson (1967).

TABLE 5. Paleomagnetic data for Precambrian formations studied in India (symbols as in Table 3)

Site	Geological formation and locality	No. of rocks or specimens	Age (my)	D°	I°	α_{95}°	Pole position λ_p	Pole position L_p	d_p°	d_m°	Reference
1.	Malani Rhyolites	170	745	353	+56	10	78°N	45°E	10.3	14.4	Athavale *et al.* 1963; Crawford and Compston 1970
2.	Rewa Sandstones	13	800	32	−37	13.7	35°N	138°W	9.6	15.9	Athavale *et al.* 1972 and Crawford 1969
3.	Mundwara Complex	25	650–850	329	−24	21	42°N	65°W	12.0	22.4	Athavale 1963
4.	Upper Vindhyan sandstones, Kaimur series	218	910–940	0	+30	6	82°N	103°W	3.7	6.7	Mishra 1965; Turgarinov *et al.* 1965
5.	Veldurti Hematites	170	Precambrian	133	−37	14	45°N	27°W	9.6	16.4	Verma *et al.* 1966
6.	Cuddapah sandstones	24	1400–1600	294	− 6	8	22°N	20°W	4.0	8.0	Prasad 1966
7.	Hyderabad dike	26	Precambrian	44	− 3	5	43°N	173°W	2.5	8.0	Verma *et al.* 1968
8.	Mysore dike	4	Precambrian	141	−75	10	34°N	55°E	18.5	16.3	Hasnain and Qureshy 1971

SARKAR, S. N. 1968. Precambrian stratigraphy and geochronology of peninsular India. Dhanbad, Publishers, India.

SARKAR, S. N., SAHA, A. K., and MILLER, J. A. 1969. Geochronology of the Precambrian rocks of Singhbhum and adjacent regions, Eastern India. Geol. Mag., **106**, pp. 15–45.

STRANGWAY, D. W. 1965. Interpretation of the magnetic anomalies over some Precambrian dykes. Geophysics, **30** (5), pp. 785–796.

——— 1966. Rock magnetism and geological correlation. Mining Geophysics, **1**, SEG Publ., pp. 54–66.

VERMA, R. K., PULLAIAH, G., and BHALLA, M. S. 1966. The paleomagnetism of the hematites from Veldurti, Cuddapah basin, India. Geophys. J. R. astr. Soc., **11**, pp. 499–505.

VERMA, R. K., PULLAIAH, G., and HASNAIN, I. 1968. Paleomagnetic study of a dyke near Hyderabad. Bull. N.G.R.I., 6, pp. 79–86.

VINGRADOV, A. P. and TUGARINOV, A. I. 1967. Geochronology of the Indian Precambrian. XXII Int. Geol. Cong., Montreal, Canada.

WASHINGTON, H. S. 1922. Deccan traps and other plateau basalts. Geol. Soc. Am., Bull. 33, pp. 765–803.